Modeling Visualization as an Affective Trigger for Human Decision-Making: The VP-RAFIAH Framework

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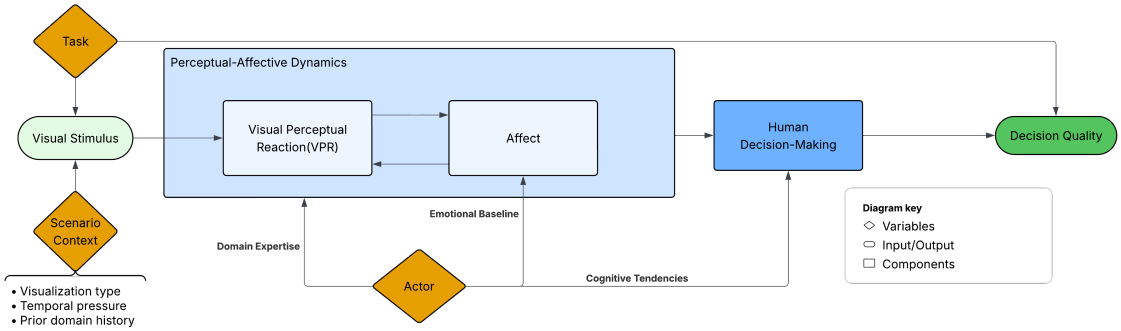


Figure 1: The VP-RAFIAH Framework and its core components, variables, and decision quality outcome.

Abstract

Visualizations support decision-making, but the steps between seeing a chart and acting on it remain opaque. No formal model captures how seeing moves through emotion into decision quality, so there's no way to measure it or design against it. VP-RAFIAH (Visual Perceptual Response-Affect Framework for Integrated Affect and Human Decision-Making) addresses this gap by modeling how a visual perceptual response (VPR) generates an affective state, how that state interacts dynamically with ongoing perceptual processing, and how this bidirectional relationship ultimately determines decision quality. Situated in the visualization domain, the framework integrates Ekman's basic emotion taxonomy with Russell's valence-arousal circumplex. It frames the full perceptual-affective-decision pipeline around three modulating variables: Task, Scenario Context, and Actor.

Index Terms: Visualization, Affect, Decision-Making, Human-Computer Interaction, Visual Perception, Emotion

1 Introduction

Visualization research has long established that effective visual design supports human decision-making [5]. The underlying mechanism remains unclear. What happens between the moment an actor sees a chart and the moment they act on it? Visualizations are not neutral conduits. They are designed tools that encode information through color, shape, spatial arrangement, and scale. These are properties that can carry an emotional signal alongside an informational signal. A red alert gauge and a green stability line can represent identical thresholds yet produce opposite affective responses. Those are not incidental to the decision but are part of how the decision is made. Existing frameworks in HCI address either the perceptual properties of visualizations [10] or model the visualization-to-decision pipeline as dual-process reasoning [5], leaving affect implicit. Without a model, two actors read the same chart, decide differently, and nothing tells us where the divergence entered, or whether emotion sharpened the read or distorted it. Naming the pathway makes it measurable and actionable, enabling personalized design. VP-RAFIAH fills that gap, modeling the affective pathway that the cognitive account brackets. It models how a visual perceptual response (VPR) generates an affective state, how that state feeds back into ongoing perceptual processing, and how this bidirectional dynamic shapes decision quality. It integrates Ekman's basic emotion taxonomy [2] with Russell's valence-arousal circumplex [7] to represent affective states, and situates the full perceptual-affective-decision pipeline within three modulating variables: Task, Scenario Context, and Actor.

2 Framework Components

VP-RAFIAH has four main components: **Visual Perceptual Response (VPR), Affect, Perceptual-Affective Dynamics, and Human Decision-Making (HDM).** Each component in VP-RAFIAH is a structural stage in the pipeline. It is a discrete mechanism through which the actor engages with the visualization. **VPR** is an immediate, pre-cognitive response to a visual stimulus, and comes from an actor's experiences [3], including cultural associations like seeing a rival's colors and reacting negatively. VPR serves as the initial perceptual signal that initiates the downstream affective state. A novice and an expert encountering the same stimulus are an identical VPR yet arrive at entirely different affective outcomes, a divergence that the framework accounts for through the Actor variable. **Affect** separates from VPR due to the cognitive nature of emotions [6]. Derived from the state produced by VPR, affect operates bidirectionally. Someone's emotional state can influence their VPR, which in turn influences their emotions. Combining Ekman's basic emotions [2] with Russell's valence-arousal spectrum [7] provides a quantitative metric for affect, rather than treating them as isolated states, thereby capturing their dynamic and changeable nature. **Perceptual-Affective Dynamics** refers to the interplay between an actor's VPR and affective state. It is not a clean one-way sequence where perception → emotion → decision. The actor's affective state shapes perceptual attention, which in turn modifies the affective state, thereby reshaping the attention again. This also interacts with the nature of the visual. A simple bar graph places low

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perceptual demand on the actor, leaving affect to do more of the processing. In contrast, a decision-making process interacts with these factors. **HDM** refers to the actor's decision-making process as shaped by the three previous factors. We measure decision quality over outcome, as a decision can produce a good result for the wrong reasons and a bad result for the right ones. A clinician who reads a live EEG monitor visually correctly, makes every call the data supports, and loses the patient has made a quality decision. Someone who misreads a chart, acts in panic, and, coincidentally, still saves a patient has not. Outcome is what happened, while quality is whether the process that produced it was sound, holistic, and replicable across contexts, and more directly tied to the framework's components than any specific outcome would be.

3 Variables

VP-RAFIAH operates across three variables: Task, Scenario Context, and Actor, which are not separate inputs, but lenses that shape how the four components play out in any given situation. Variables are distinct from components, as components describe what happens in the pipeline, while variables describe the conditions under which it happens.

Task refers to what the actor is trying to accomplish when engaging with the visual. Tasks range from low-stakes decisions, like using Google Maps to pick a route, to high-stakes judgments, like a surgeon interpreting a scan. Complexity and stakes don't always move together. What matters is how the task conditions the perceptual-affective dynamics. A time-pressured, high-stakes task compresses the cycle, leaving fewer opportunities for perceptual-affective feedback before a decision must be made [4]. The Task also defines what a quality decision looks like in context.

Scenario context encompasses the full environment in which the actor interacts with the visual. It's always a visualization in this framework, since VPR requires a visual stimulus to initiate the pipeline. Context includes temporal pressures, the engagement setting, and the actor's prior history with similar visuals or domains. The same chart in a calm research setting and under deadline pressure represents two entirely different contexts. Context is not static, as it shapes and is shaped by the actor's VPR and affective state throughout the process.

The **actor** is the human individual engaging with the visual. At least three factors shape their experience: (1) *emotional baseline*, where the actor sits on Russell's spectrum [7] before encountering the visual, which conditions the affective state VPR feeds into; (2) *cognitive tendencies* [1], predispositions like impatience that cause actors to misread visual information or anxiety that cause actors to overread (attributing more certainty or signal to the visual than the data supports) from a lack of confidence, skewing the perceptual-affective dynamic [9]; and (3) *domain expertise*, accumulated experience that shifts the balance of perceptual-affective dynamic toward VPR and away from affective override.

4 Methodology

VP-RAFIAH was developed through an iterative conceptual modeling process. An initial systematic literature review found no existing framework that connects visual perception, affect, and decision quality, motivating the Development of VP-RAFIAH. We conducted a broader survey spanning HCI, visualization research, cognitive psychology, and decision-making theory. Established models like Ekman's basic emotion taxonomy [2], Russell's valence-arousal circumplex [7], Ware's pre-attentive visualprocessing theory [10], and Van Wijk's model of visualization value [8] were synthesized to ground the framework's four components.

5 Working Example

The value VP-RAFIAH brings is clearest in what it can differentiate. Consider the following example: Two actors observe the same cascading series of red candlesticks on a live stock chart during sharp intraday volatility. The newer investor registers red as danger, their affect shifts toward panic, and they place a reactive sell order as their perceptual-affective dynamic never has a chance to balance out. The experienced hedge fund trader sees the same stimulus, immediately contextualizes it as a known consolidation pattern, and stays alert rather than afraid, choosing to hold or buy into the position. Same visual stimulus, divergent decision quality driven entirely by differences in actor variables: domain expertise and emotional baseline. A newer investor who sells and coincidentally avoids further decline has not made a quality decision. The process was affectively compromised regardless of the result. The outcome is a single draw. Quality is what holds across draws, and a process that panics on red will misfire the next time it appears. The lucky sell is a losing strategy iterated. VP-RAFIAH locates the divergence in the Actor variable, domain expertise, and emotional baseline; traces how it propagated through compressed perceptual-affective dynamics and VP's affective override; and specifies what a quality decision required: a stable affective baseline, a calibrated VPR, and an uncompressed iterative processing cycle. Without the framework, both decisions are just choices. With it, one is structurally sound, and one is structurally compromised, and the framework tells us exactly why, in terms precise enough to operationalize, measure, and eventually design against.

6 Next Steps

Our next steps include a formal evaluation of the framework across additional domains, operationalizing decision-quality measurement in empirical contexts, and expanding the framework to multi-actor scenarios in which group dynamics and interpersonal affect may influence the VPR-HDM pipeline.

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